11 **(a)** State the condition required for a protostar to become a stable star.

.....

.....

..... [1]

(b) **(i)** Define the Hubble constant.

.....

.....

..... [2]

(ii) The current estimate for the Hubble constant is 2.2×10^{-18} per second.

State the equation which gives an estimate for the age of the Universe.

..... [1]

(iii) Calculate an estimate for the age of the Universe.

estimate of age = ... s [1]

[Total: 5]

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